31N0404

Roll No. _____ Total No. of Pages.: 04

31N0404

B.Tech III Sem Main/Back Exam 2025-26

Computer Science and Engineering and Allied Branches

(3CS4-04) Object Oriented Programming Using C++

3AD4-04, 3AM4-04, 3AI4-04, 3CM4-04, 3CY4-04, 3CD4-04, 3CS4-04, 3IT4-04, 3IC4-04, 3VD4-05 - Common to CS, IT, AI, DS,MC, CM, CD, CA, AD, AM, CY, IO, EE(VLSI Design & Technology)

Time : 3 Hours **Maximum Marks: 70**
Min. Passing Marks:

Instructions to Candidates:

Part-A: Short answer questions (up to 25 words) 10x2 marks = 20 marks. All ten questions are compulsory.

Part- B: Analytical/Problem Solving questions 5x4 marks = 20 marks. Candidates have to answer 5 questions out of 7.

Part – C: Descriptive/Analytical/Problem Solving questions 3x10 marks = 30 marks. Candidates have to answer 3 questions out of 5.

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting materials is permitted during examination. (Mentioned in form No. 205)

1_____ 2_____

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Part-A

[2x10=20]

Q.1. What are the core principles of OOP?

Q.2. List various applications of OOP.

Q.3. How Static Variable is different from Static member.

Q.4. Explain the working of Hierarchical Inheritance.

Q.5. Explain the concept of Virtual Function.

Q.6. Explain the concept of Late Binding.

Q.7. What is the ambiguity in Function Overloading?

Q.8. Explain the concept of Pure Virtual Function?

Q.9. Explain the concept of Copy Constructor.

Q.10. Explain the concept of user defined namespaces.

PART-B

[5x4=20]

Q.1. Explain public, private and protected access specifiers and show their visibility when they are inherited as public, private and protected.

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Q.2. With the help of a program explain, what is the difference between opening a file with constructor function and opening a file with open ( ) function.

Q.3. What is encapsulation and why is it important? How does it help in data hiding and information security?

Q.4. How does polymorphism enhance the flexibility and maintaining of object-oriented systems? Discuss its role in dynamic binding.

Q.5. Which operates cannot be overloaded? Write steps to overload + operator so that it can add two complex numbers.

Q.6. Why is type casting? What are explicit and implicit type conversions?

Q.7. What is Standard Template Library ? How is it different from the C++ Standard Library?

PART- C **[3x10=30]**

Q.1. What does inheritance means in c++? What are different forms of inheritance? Give an example of each.

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Q.2. With the help of an example differentiate between method overloading and method overriding.

Q.3. Write a C++ program demonstrating use of the pure virtual function with the use of base and derived classes.

Q.4. Write a C++ program involving the handling of exceptions in constructors and destructors.

Q.5. Discuss the role of designs patterns in buildings robust and scalable object-oriented systems.

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